

CHARGING & DISCHARGING OF CAPACITOR

OBJECT:- To study the Charging & Discharging of a capacitor and to find out the time constant.

APPARATUS: Charging & Discharging of capacitor experimental set up.

FORMULA/THEORY:-

a). In case of charging the voltage across capacitor is given by following equation

$$V = V_0 \left(1 - e^{-\frac{t}{RC}} \right)$$

When $t = RC = \tau = \text{time constant}$, then

$$V = V_0 (1 - e^{-1}) = V_0 \left(1 - \frac{1}{e} \right) = 0.632 V_0$$

So, in case of charging, the time constant is the time when voltage across capacitor increases from zero to $(1 - e^{-1})$ i.e., 0.632 times its maximum value.

b). In case of discharging the voltage across capacitor is given by following equation

$$V = V_0 e^{-\frac{t}{RC}}$$

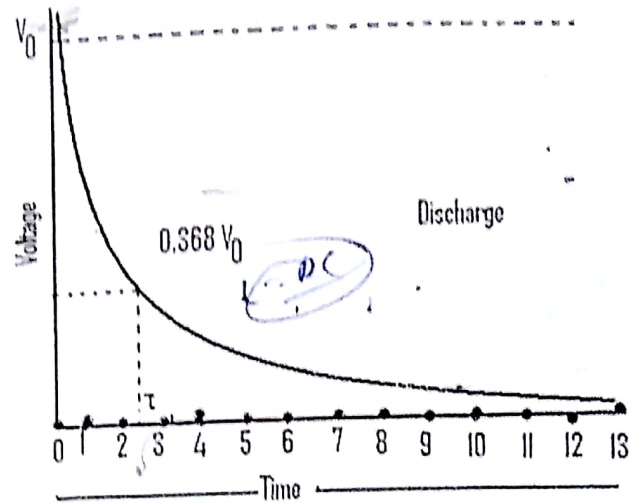
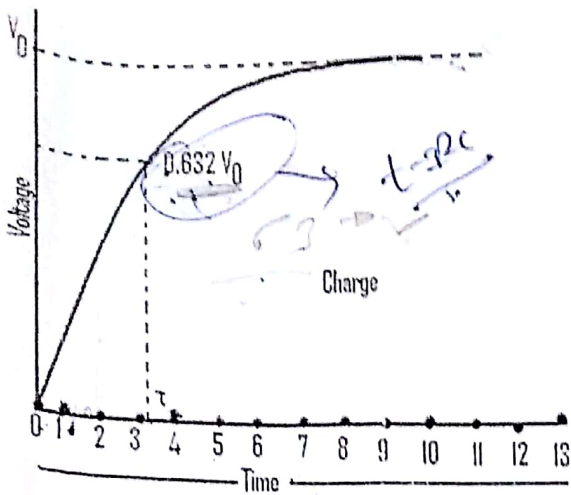
When $t = RC = \tau = \text{time constant}$, then

$$V = V_0 e^{-1} = V_0 \frac{1}{e} = 0.368 V_0$$

So, in case of discharging, the time constant is the time when voltage across capacitor decays to e^{-1} i.e., 0.368 times its maximum value.

DIAGRAMS:-

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GENERAL INSTRUCTIONS:-

The main property of capacitor is its ability to store electrical charge. When a voltage source is connected to capacitor through a high resistance, it begins to charge. Time taken by capacitor to charge 63% of applied voltage or discharge by 37% of initial voltage is called "Time Constant". Theoretical value of time constant is RC seconds, where R is in Ohms and C in farads.

The experiment for study of charge and discharge of a capacitor is very simple to perform. The built circuit consists of following:

- Built-in stabilized D.C. power supply of 1.2 to 10 volts.
- High resistance voltmeter having internal resistance of 25K Ohm/volt.
- High resistance of 10K, 15K and 18K Ohms and capacitor of 1000mfd, 2200mfd and 4700mfd.
- Toggle switches for charging and discharging capacitor through high resistance.
- Dump switch is provided to short circuit capacitor and bring back the meter pointer to zero quickly.

OPERATING INSTRUCTIONS:-

1. Make sure that pointer of voltmeter is at zero mark. In case it is indicating some reading above zero, bring the pointer to zero by pressing dump switch. By doing so capacitor will be discharged if any charge was present initially. Further bring the pointer at zero by adjusting zero adjustor provided on voltmeter.
2. Connect 'X' to any one of the high resistance R_1 , R_2 and R_3 connect Y to any one of the capacitances C_1 , C_2 or C_3 . Choose $R_1 C_1$ such that time constant RC is appreciable.
3. Press the charge switch for charging capacitor through resistance and note down the reading of voltmeter after every 5 to 10 seconds repeating same procedure till capacitor is almost fully charged. See the example after the observation table. It is better to take the time corresponding to changing voltmeter reading ($0V_1$, $1V_1$, $2V_1$)
4. When capacitor attains full charge. Bring the switch towards discharge and note down voltmeter reading of voltmeter every 5 to 10 seconds till the discharged.
5. Now plot curves for charge and discharge taking suitable scale. Time is plotted on X-axis while Voltage on Y-axis.
6. Measure the input voltage V_0 . since you know it you don't need to charge the capacitor fully to find it.

DETERMINATION OF TIME CONSTANT:-

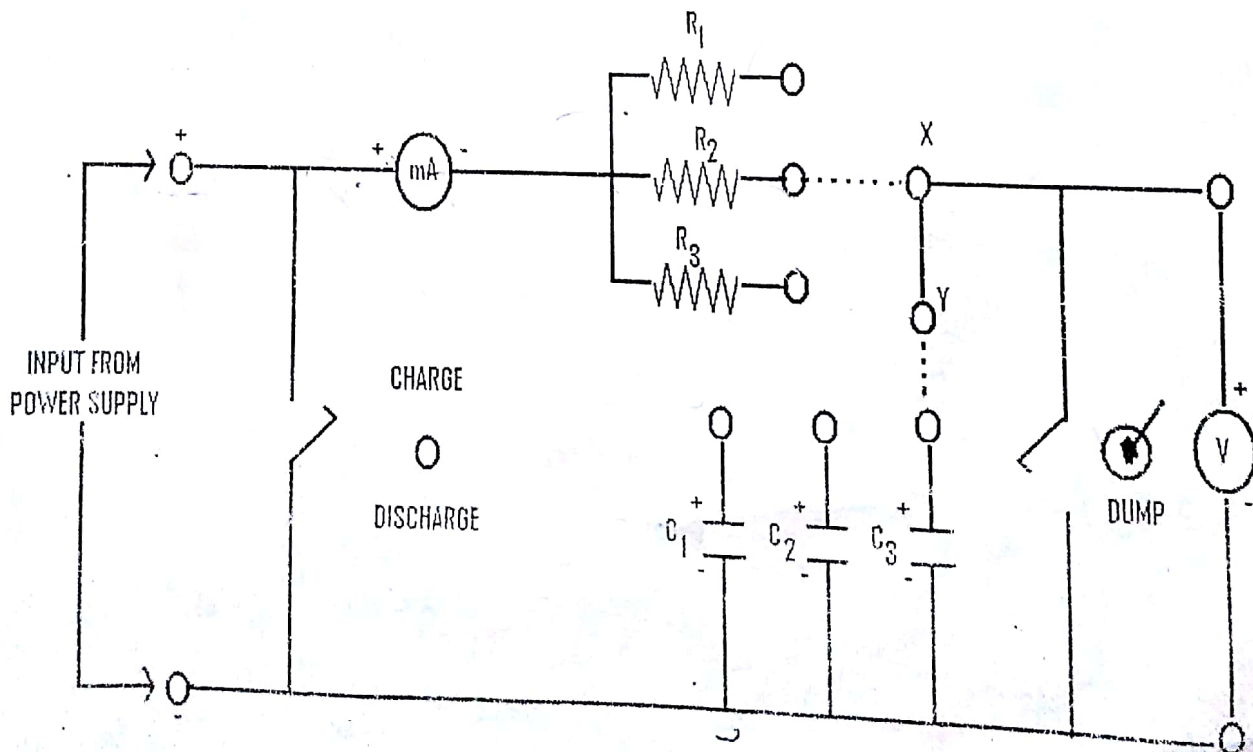
Suppose $4700\mu\text{fd}$ i.e. 4700×10^{-6} farad capacitor is charged through a 10000 Ohm resistor. The voltage on capacitor shall reach 63% of applied voltage after CR seconds (Time Constant) i.e. $4700 \times 10^{-6} \times 10^4$ or 47 seconds. The value of time constant may also be verified by charge and discharge curves. Suppose highest voltage across capacitor is 10 volts. The time taken to charge it upto 6.3 V (63% of 10

volts) may be determined by drawing a straight line parallel to X-axis at 6.3V and find out corresponding value of time. Similarly during discharge value of time constant may be find out by drawing a line parallel to X-axis at 6.3V (37% loss of 10 Volts or 6.3V).

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NOTE:-

1. Ammeter of 1-0-1 mA is provided just to observe the current direction while charging and discharging of the capacitor. It will be noted that current direction is opposite to each other while charging & discharging.
2. While conducting the experiments (Charging & Discharging) don't dump switch towards dump side only.

DIAGRAM:-

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OBSERVATION TABLE:-

Applied Voltage = ...1.2-10V... Volts

Resistance R = ...10... K Ohm

Capacitor C = ... μ FaradC₁, C₂, C₃

S. No.	Charging		Discharging	
	Voltage (Volts)	Time (second)	Voltage (Volts)	Time (second)
1.	1 2 3	3.72 5	10	
2.	2 4 6	7.62	9	
3.	3 6 9		8	30
4.	4 9 14	12.69 10.2	7	
5.	5 11 22		6	
6.	6 15.9 32	31.95 21.8	5	
7.	7 20.4		4	2.14
8.	8 28.9		3	
9.	9 38.9		2	2.14 2.26
10.	10 45.2		1	2.20
11.				
12.				

Example 1: $R = 10K\Omega$, $C = 4700\mu f$

$$RC = 10 \times 10^3 \times 4700 \times 10^{-6} = 47 \text{ seconds}$$

Example 2: $R = 15K\Omega$, $C = 4700\mu f$

$$RC = 15 \times 10^3 \times 4700 \times 10^{-6} = 70.5 \text{ seconds}$$

(Note: Always choose V_0 such that $0.63V_0$ is appreciable and the reading between the marking is not admissible. If at $t = 10$ seconds the voltmeter needle is between 2 and 2.4 then the reading is 2 and the error is very large. So note the time as the voltage changes.

In example 1: $V_0 = 10$ volts take reading for

$$V_c = .8, 1.6, 2.4, 3.7, 4.0, 4.8, 5.6, 6.4, 7.2, 7.6, 8.0, 8.4, \dots, 9.6$$

Each division on voltmeter is 0.4 volts. Since till $t = t_c$ the graph is almost linear and you are taking 8 reading to get 6.4 volts. You will be taking a reading $(47/8 \approx 6)$ approximately every 6 seconds.

There are two ways determine V_0 (i) wait long enough for capacitor to charge fully (ii) use the multimeter to measure the voltage of the battery, then you don't have to wait for the capacitor to charge fully.)

RESULT:-

The time constant from graph =Second

Theoretical value of time constant =Second

PERCENTAGE ERROR:-

PRECAUTIONS:-

1. Condensers should not be leaky.
2. Appropriate value of R and C given. quite a good number of observations should be used.
3. There should be carefully connections.

VIVA - VOCE

Q.No.01 The time of charging of a capacitor having higher value through a same resistance will be smaller or longer?

Q.No.02 The time taken for the charging of a capacitor increases or decreases if the resistance in the circuit is increased?

Q.No.03 What is the voltage across the capacitor when it is charged fully?

Q.No.04 What is the current in the circuit after the capacitor is fully charged?

Q.No.05 What is the potential difference across the resistance when the capacitor is fully charged in a charging circuit?

Q.No.06 What is the time constant for the above circuit?

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